A failure of complete monotonicity for a Mills-ratio quotient

Research proof and Lean formalization
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Abstract

We refute Ferreira–Simon Conjecture 3 on quotients of derivatives of the Gaussian Mills ratio. Already the quotient $-r^2/r'$ has a negative fortieth derivative throughout some interval of positive inputs. The proof reduces the derivative at zero to an integer polynomial in $\pi/2$ of degree 21. An exact rational Bernstein certificate proves its negativity using six-decimal bounds on π; continuity gives a regular positive-input counterexample. The literal Gaussian tail, derivative recurrences, finite polynomial certificate, and positive-interval conclusion are formalized in Lean.

1 The conjecture and the counterexample

The Gaussian Mills ratio is

$$r(x) = e^{x^2/2} \int_x^\infty e^{-t^2/2}\, dt, \qquad x \in \mathbb{R}. \tag{1}$$

A function g is completely monotone on $(0,\infty)$ if it is smooth there and $(-1)^k g^{(k)}(x) \geq 0$ for every integer $k \geq 0$ and every $x > 0$. Ferreira and Simon [1], Conjecture 3 in arXiv v2, p. 31, conjecture that

$$x \longmapsto -\frac{(r^{(n)}(x))^2}{r^{(2n+1)}(x)}$$

is completely monotone for every nonnegative integer n. We use $n = 0$. The differential equation $r' = xr - 1$ gives

$$F(x) = -\frac{r(x)^2}{r'(x)} = \frac{r(x)^2}{1 - xr(x)}. \tag{2}$$

Theorem 1. *There is a $\delta > 0$ such that, for every $0 < x < \delta$,*

$$r'(x) < 0 \qquad \text{and} \qquad F^{(40)}(x) < 0.$$

Consequently F is not completely monotone on $(0,\infty)$, and Ferreira–Simon Conjecture 3 is false.

The denominator is nonzero at every input in this conclusion. The result concerns the source's uncomposed quotient. Their Proposition 10 establishes complete monotonicity after composition with $x \mapsto \sqrt{x}$.

2 An exact derivative certificate

The Gaussian integral and differentiation of (1) give

$$r(0)^2 = \pi/2, \qquad r'(x) = xr(x) - 1. \tag{3}$$

Write $c = r(0)$, $R_j = r^{(j)}(0)$, and $D_j = F^{(j)}(0)$. The first identity in (2) is regular near zero because $r'(0) = -1$. Differentiating (3) and $F - xrF = r^2$ gives the finite recurrences

$$R_0 = c, \quad R_1 = -1, \quad R_{j+2} = (j+1)R_j, \qquad D_0 = c^2, \tag{4}$$

$$D_{n+1} = \sum_{j=0}^{n+1} \binom{n+1}{j} R_j R_{n+1-j} + (n+1) \sum_{j=0}^{n} \binom{n}{j} R_j D_{n-j}. \tag{5}$$

These are polynomial identities over $\mathbb{Z}[c]$. Iterating (5) to $n + 1 = 40$ gives

$$D_{40} = P(c^2), \qquad P \in \mathbb{Z}[T], \qquad \deg P = 21. \tag{6}$$

For orientation, the leading coefficient is 40! and numerical evaluation gives

$$P(\pi/2) \approx -2.4628065653022778370 \times 10^{22}.$$

We certify the sign by rational arithmetic rather than by this decimal. Equations (4)–(5) specify every coefficient of P; the accompanying data gives its complete ascending coefficient list and the 41 intermediate derivative polynomials.

Set

$$a = \frac{392699}{250000} = 1.570796, \qquad b = \frac{3141593}{2000000} = 1.5707965, \qquad h = b - a.$$

The bounds $3.141592 < \pi < 3.141593$ imply $a < \pi/2 < b$. If $P(T) = \sum_{j=0}^{21} A_j T^j$, define the rational numbers

$$V_i = h^i \sum_{j=i}^{21} \binom{j}{i} A_j a^{j-i}, \qquad B_k = \sum_{i=0}^{k} V_i \frac{\binom{k}{i}}{\binom{21}{i}} \quad (0 \le k \le 21). \tag{7}$$

The power-to-Bernstein identity is

$$P(a + hu) = \sum_{k=0}^{21} B_k \binom{21}{k} u^k (1-u)^{21-k}. \tag{8}$$

Exact evaluation of the finite rational expressions in (7) gives, for each of the 22 indices,

$$-2463 \cdot 10^{19} < B_k < -2460 \cdot 10^{19} < 0. \tag{9}$$

The retained certificate stores both B_k and $C_k = B_k\binom{21}{k}$. Lean checks the full identity (8) with the C_k and checks $C_k < 0$ for every index. Since $\binom{21}{k} > 0$, these sign checks also give $B_k < 0$. The stronger rounded bounds (9) are independently checked by exact rational arithmetic in the manuscript QA record.

Take $u = (\pi/2 - a)/h \in (0, 1)$. Every summand in (8) is strictly negative, so $P(\pi/2) < 0$. By (3) and (6),

$$F^{(40)}(0) < 0.$$

To finish Theorem 1, the Gaussian tail is smooth on $\mathbb{R}$, and the denominator of (2) is 1 at zero. Thus $F^{(40)}$ is continuous near zero. Its strict negative value persists on a neighborhood, as does $r'(0) = -1 < 0$. Choose $\delta > 0$ inside both neighborhoods. For every $0 < x < \delta$, the even-order complete-monotonicity inequality fails at order 40. This proves the theorem on genuinely positive inputs.

3 Formalization, reproducibility, and scope

The Lean certificate defines r by the literal improper Gaussian integral in (1). It proves the finite-integral bridge, the initial value, the differential equation, and smoothness. Finite Leibniz identities link the 41 archived integer polynomials to the genuine iterated derivatives of (2); all table identities are proved. The Bernstein identity and signs are then checked exactly. The endpoint

`MillsCounterexample.source_counterexample_351688`

is a premise-free negation of the literal source conjecture. The stronger interval endpoint also proves that r' stays negative there. There is no residual differential-equation, polynomial-identity, or numerical-bound assumption in these endpoints.

The certificate has 47 declarations in seven modules, built against Lean 4.34.1 and mathlib `d13f23b7`. The replay procedure copies source files afresh, checks pinned dependencies, treats warnings as errors, inspects every declaration's type and axioms, and excludes proof holes, extra axioms, and native decision procedures. The exact verification script independently reconstructs the recurrences and Bernstein identity. A separate Cauchy quadrature diagnostic of the complex error-function formula agrees with the derivative value to relative error below 4×10^{-83}; this numerical diagnostic is not used in the proof.

This is one refutation, already at $n = 0$. We make no assertion about the least derivative order of failure or the remaining individual values of n. The source comparison uses the latest arXiv v2; the published full-text access limit is recorded. The bounded prior-work investigation and independent review remain pending for this draft, and no absolute priority claim is made. Later complete-monotonicity results such as [2] concern different weighted or composed ratios.

References

[1] R. A. C. Ferreira and T. Simon, *Convolution of beta prime distribution*, Trans. Amer. Math. Soc. **376** (2023), 855–890. doi:10.1090/tran/8748; arXiv:2108.09244v2.

[2] Z.-H. Yang, *Extension of a complete monotonicity theorem with applications*, arXiv:2507.10954 (2025).